MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY,
TTL, COUNTERS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/327A, dated 10 December 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, line 2: Delete "Three" and substitute "Two".

1.2.3, outline letter: Add "2" case outline: Opposite "2", add "C-2 (20-terminal, .350" x .350") square chip carrier package.

1.3, thermal resistance, delete and substitute the following:

"A, B, D, and F	70°C/W
C and E	50°C/W
2	60°C/W"

PAGE 2

1.4, setup time (types 01 and 02): Delete "Clear inactive static setup time" and substitute "Clear inactive state setup time".

1.4, last line: Delete "25 ns minimum" as added by previous amendment, and substitute "35 ns minimum".

PAGES 3 AND 4

TABLE I, under "Conditions" heading: Add "$-55°C \leq T_C \leq +125°C$".

PAGE 4

* TABLE I, propagation delay times, low to high (t_{PLH2}) and high to low (t_{PHL2}), device 02, maximum limits column: Delete "84" and substitute "93".

PAGE 5

TABLE II: Delete class C devices column in its entirety.

3.6, delete and substitute:

"3.6 Marking. Marking shall be in accordance with MIL-M-38510."

FSC 5962

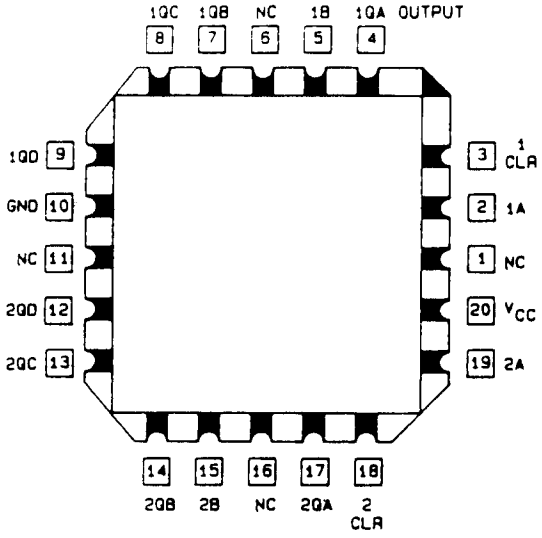
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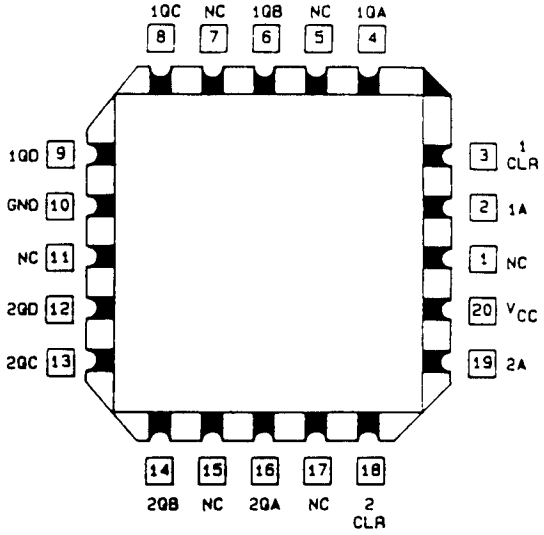
FIGURE 1: Add terminal connection for case 2 for device types 01 through 03, as printed below:

"

Device type 01
Case 2



Device type 02
Case 2



Device type 03
Case 2

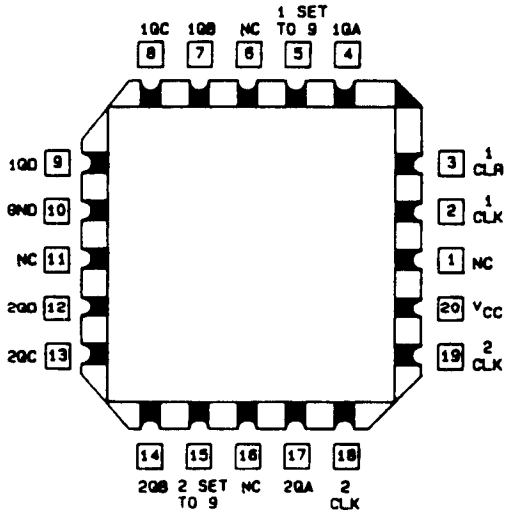


FIGURE 1. Terminal connections (top view) - Continued."

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PAGE 12

FIGURE 5, note 1: Add $t_{p(clear)} \geq 35$ ns for device 02.

PAGES 16, 17, AND 18

TABLE III, device 01, under cases: Delete pin designations and substitute the following:

"

Cases	Pins not referenced are NC															
E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	1A	1CLR	1QA	1B	1QB	1QC	1QD	GND	2QD	2QC	2QB	2B	2QA	2CLR	2A	V_{CC}

"

* TABLE III, device 01, subgroup 1, test number 28: Delete "0.4 V" under pin 13. Add "0.4 V" under pin 12.

PAGE 17

* TABLE III, device 01, subgroup 9, test number 76, symbol column: Delete "t_{PLH2}" and substitute "t_{PHL2}".

PAGE 18

* TABLE III, device 01, footnote 8: Delete "A = 2.0 V min. and B = 0.7 V max." and substitute "A = 2.4 V and B = 0.4 V".

PAGES 19, 20, AND 21

TABLE III, device 02, under cases: Delete pin designations and substitute the following:

"

Cases	Pins not referenced are NC													
A,B,C, D,E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14
2	2	3	4	6	8	9	10	12	13	14	16	18	19	20
	1A	1CLR	1QA	1QB	1QC	1QD	GND	2QD	2QC	2QB	2QA	2CLR	2A	V_{CC}

"

PAGE 20

* TABLE III, device 02, subgroup 9, test numbers 82 through 85 (t_{PLH2} and t_{PHL2}), maximum limits column: Delete "65" and substitute "70".

* TABLE III, device 02, subgroup 10, test numbers 82 through 85 (t_{PLH2} and t_{PHL2}), maximum limits column: Delete "84" and substitute "93".

PAGE 21

* TABLE III, device 02, subgroup 11, test numbers 82 through 85 (t_{PLH2} and t_{PHL2}), maximum limits column: Delete "84" and substitute "93".

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Footnote 3, delete and substitute:

"3/ I_{IL} limits shall be as follows:

Test	Min/Max limits (mA)				
	Circuits				
	A	E	B	D	F
I_{IL1}	-0.15/-0.38	-0.12/-0.36	-0.16/-0.40	-0.12/-0.36	-0.12/-0.36
I_{IL2}	-0.35/-1.6	-1.0/-2.4	-0.35/-1.60	-0.35/-1.60	-0.35/-1.60

"

* TABLE III, device 02, footnote 4: Delete "A ≥ 2.0 V and B ≤ 0.7 V" and substitute "A = 2.4 V and B = 0.4 V".

PAGES 22, 23, AND 24

TABLE III, device 03, under cases: Delete pin designations and substitute the following:

"

Cases	Pins not referenced are NC															
E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20
	1CLK	1CLR	1QA	1ST9	1QB	1QC	1QD	GND	2QD	2QC	2QB	2ST9	2QA	2CLR	2CLK	V_{CC}

"

PAGE 24

* TABLE III, device 03, footnote 7: Delete "A = 2.0 V min. and B = 0.7 V max." and substitute "A = 2.4 V and B = 0.4 V".

PAGE 25

4.2c, delete and substitute:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

4.4.3b: Delete in its entirety.

PAGE 26

4.4.5: Delete in its entirety.

6.1: Delete in its entirety.

6.4: Delete "Electrical ground (common terminal) and substitute "Ground zero voltage potential".

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6.7, delete and substitute:

"6.7 Manufacturers' designations. Manufacturers' circuits included in this specification are designated as shown in table IV.

TABLE IV. Manufacturers' designations. 1/

Device type	Manufacturers				
	Texas Instruments	Signetics	Motorola	Fairchild	National Semiconductor
	Circuit A	Circuit B	Circuit D	Circuit E	Circuit F
01	X	X		X	
02	X	X	X	X	X
03	X	X		X	

"

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1250)

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